PHASIC ALERTNESS CUES MODULATE VISUAL PROCESSING SPEED IN HEALTHY AGING

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CONCLUSIONS

Phasic auditory alertness cues lead to an increase in visual processing speed

Alerting effects are preserved in healthy aging

SOA effects propose inverse U-shaped function of alerting and processing speed

FORMER RESEARCH

Visual and auditory alertness cues increase visual processing speed
Matthias et al., 2010; Peterson et al., 2017

Visual processing speed declines over the lifespan
McAvinue et al., 2012

Speeded paradigms: extrinsic alerting effect decreases in healthy aging
Festa-Martino et al., 2004

No significant extrinsic alerting benefits on visual attention capacity in healthy aging
Wiegand et al., 2017

Temporal orienting of attention preserved in elderly
Chauvin et al., 2016

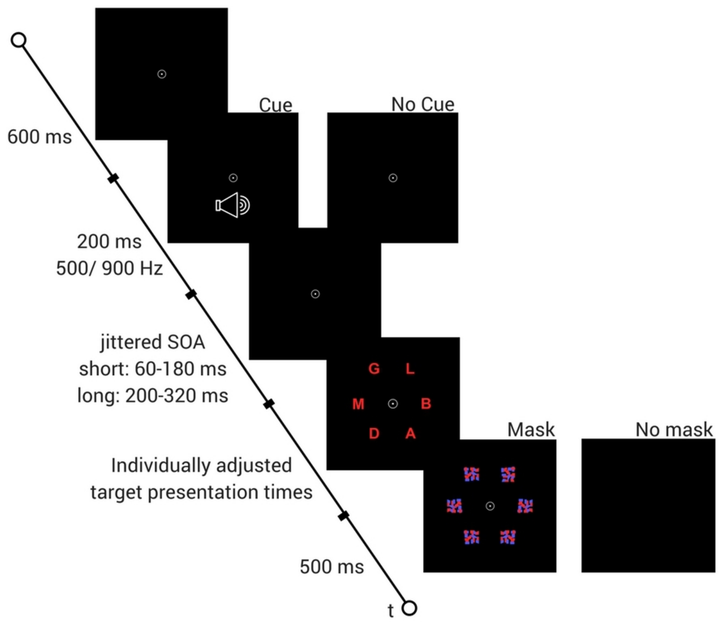
Are phasic alerting effects on visual processing speed preserved in healthy aging?

PARTICIPANTS

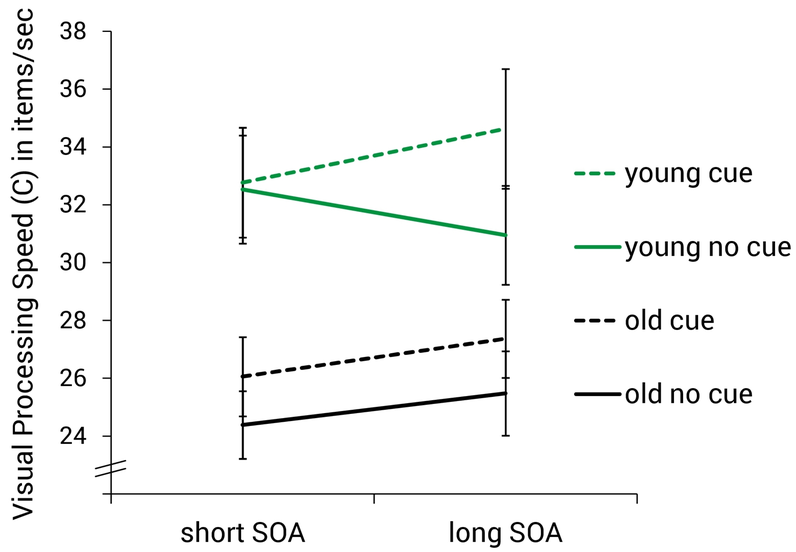
	Young n=34	Old n=32
mean age (SD)	26.4 (4.7)	71.9 (4.7)
sex (female/male)	21/13	11/21
handedness (r/l/ambidex.)	29/5/0	30/0/2
mean education in y (SD)	12.6 (0.9)	11.8 (1.8)
MWTB score (SD)	28.6 (4.5)	32.7 (2.2)
MMSE score (SD)	/	28.8 (1.1)

WHOLE REPORT PARADIGM

based on the Theory of Visual Attention (Bundesen, 1990)

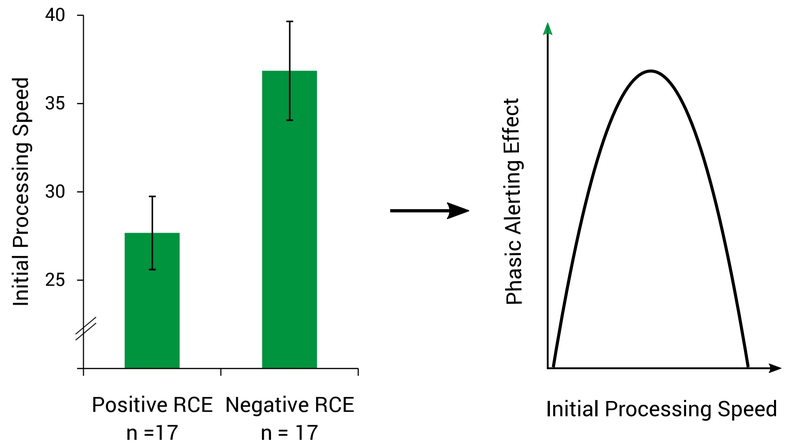


PHASIC ALERTING EFFECTS



Cue	$p = .007$, $BF_{10} = 4.759$	$p = .010$, $BF_{10} = 1.417$
SOA	$p = .823$, $BF_{10} = 0.182$	$p = .266$, $BF_{10} = 0.453$
Cue x SOA	$p = .040$, $BF_{10} = 3.000$	$p = .887$, $BF_{10} = 0.236$

Short SOA in young participants: Relative Cueing Effect (RCE)



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